DE–2849

Sub. Code
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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List any types of number system and explain with an example for each.

2. Draw truth tables for the given expression :

   (a) $x.y.z + \overline{x}\,\overline{y}\overline{z}$

   (b) $A.(BC + \overline{B}C)$

   (c) $A.(\overline{B}C + B\overline{C})$

   (d) $(A + B).(B + C).(\overline{A} + \overline{B})$

3. Explain multiplexer with block diagram and gates.

4. Discuss full adder with block diagram and gates.

5. Describe counters and its types.

6. How is clock used and where it is used? Explain.

7. Show the logic diagram of clocked D-flip flop with AND and NOR gates.

8. Discuss the generations of the computer.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss about binary arithmetic operations with examples.

10. Explain Karnaugh map simplification.

11. Briefly discuss about encoder and decoder.

12. How do you construct an ALU? Discuss.

13. Discuss shift registers and binary counters with an example.

14. Classify computer memory and describe its types with explanation.

15. Explain the following :

(a) How are data represented?

(b) Types of processor.

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain the while and do-while loops with suitable example.

2. Explain the arithmetic and relational operators with example.

3. Explain the different types of functions with suitable examples.

4. Write a program to sort the given set of numbers in ascending order.

5. Explain the array of pointers with examples.

6. Explain any four file handling functions.

7. Write about Queues and its representation.

8. Explain about doubly linked list.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Write about the different data types available in C.

(b) Explain in detail about the gets and puts functions with suitable example.

10. (a) Explain the else-if ladder with suitable example.

(b) Explain the switch statement with example.

11. (a) Write a program to find the factorial of a given number using recursive function.

(b) Write about arrays with example.

12. Explain the various preprocessor directives with example.

13. (a) Explain in detail about structures.

(b) Explain the 'malloc' and 'calloc' functions.

14. (a) Write short notes on abstract data types.

(b) Explain the various stack operations.

15. What is a binary Tree? Explain in detail about the operations on binary trees and the applications of binary trees.

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DISTANCE EDUCATION

M.Sc. (Software Engineering (5 Years Integrated)) DEGREE EXAMINATION, DECEMBER 2015.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Construct the truth table for the following formulas.

   (a) $\neg(\neg P \vee \neg Q)$

   (b) $P \wedge (P \vee Q)$

   (c) $(P \wedge Q) \vee (\neg P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$.

2. Obtain disjunctive normal forms and conjunctive normal form of each of

   (a) $P \wedge (P \rightarrow Q)$

   (b) $\neg(P \vee Q) \rightleftharpoons (P \wedge Q)$.

3. Give the power set of the following

   (a) $\{a, \{b\}\}$

   (b) $\{1, \phi\}$

   (c) $\{X, Y, Z\}$.

4. Let R be a relation on a set X and $R^{-1} = \{\langle x, y\rangle \mid \langle y, x\rangle \in R\}$ Prove that $R^{-1} = R$ if and only if R is symmetric

5. Is every relation which symmetric and transitive on a set A, always reflexive? Why or why not?

6. Show that the function $f : Z \to Z$ defined by $f(x) = \begin{cases} \frac{x}{2} & \text{when } x \text{ is even} \\ \frac{x-1}{2} & \text{when } x \text{ is odd} \end{cases}$ is primitive recursive.

7. If the binary operation $\circ$ be defined on Z, the set of all integers, by

   (a) $a \circ b = a + b + 1,\ a, b \in Z$ find the identity element with respect to $\circ$.

   (b) by $a \circ b = a + b + 1,\ a, b \in Z$, find the inverse of a with respect of $\circ$.

   (c) by $a \circ b = 2a + b$, for all $a, b \in Z$. Examine whether $\circ$ is commutative and associative.

8. Show that a simple graph is a tree if and only if it is connected, but the deletion of any of its edges produces a graph that is not connected.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Obtain the principal conjunctive normal form of the formula S given by $(\neg P \to R) \wedge (P \rightleftharpoons R)$.

(b) The truth table for the formula A is given below. Determine its disjunctive and conjuctive normal forms.

P	Q	R	A
T	T	T	F
T	T	F	F
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	T
F	F	T	F
F	F	F	T

10. Determine the following implication

(a) $((\exists x)\, P(x) \wedge Q(a)) \Rightarrow (\exists\, x) P(x) \rightarrow \neg Q(a)$

(b) $(x)\,(P(x) \vee Q\,(x)),\, (x) \neg P(x) \Rightarrow (\neg x)\, Q\,(x)$

(c) $\neg\,(x)\,(P(x) \vee Q(x)),\, (x)\, \neg P(x) \Rightarrow (x)\; Q(x)$.

11. Give a example of a relation which is

(a) Reflexive symmetric but not transitive

(b) Reflexive transitive but not symmetric

(c) Reflexive but not symmetric and transitive

(d) Symmetric but not transitive and reflexive

(e) Transitive but not reflexive and symmetric

(f) Neither Reflexive nor symmetric and transitive

(g) Reflexive symmetric and transitive

(h) Symmetric and Anti-symmetric.

(i) Anti-symmetric but not reflexive

12. Let R be a relation on $Z \times (Z \setminus \{0\})$ defined by $\langle a, b\rangle R \langle c, d\rangle \Leftrightarrow ad = bc$. Show that R is equivalence relation.

13. Let f be a function from A to B. Let S and T be subsets of B. Show that $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$ and $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

14. State and Prove the Lagrange's theorem. (With necessary results).

15. Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G. Then $r = e - v + 2$.

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the benefits of OOP.

2. Write a note on data encapsulation.

3. What is 'this' pointer? How it can be declared and used?

4. Explain compile time polymorphism.

5. Write a C++ program to overload binary operators.

6. Write a note on compile time polymorphism.

7. What is virtual function? How it works?

8. Explain the types of inheritance.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about OOP concept and its characteristics.

10. Write a note on:

    (a) Inline function

    (b) Friend function.

11. Discuss in detail about pointers and its usage in OOP.

12. Briefly explain various types of constructors. Explain how it differs from destructor.

13. Implement a program to add two complex numbers using binary operator overloading.

14. Give an example for pure virtual base class. Explain how the virtual base class can be implemented?

15. Explain multilevel and multiple inheritances with suitable example.

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DE–2853

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is an operating system? Discuss about the structure of operating system.

2. Write an overview about system calls with an example.

3. Explain in brief about policy driven scheduling and two level scheduling with examples.

4. Discuss about interrupt handlers and device drivers.

5. Write a short note on interprocess communication.

6. Discuss on multi programming with fixed partitions using relevant examples.

7. Write an overview about linked lists and buddy system with examples.

8. Explain in detail about protection mechanisms.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write an overview about various concepts of operating systems.

10. How to schedule a process? Illustrate.

11. Illustrate the steps involved in dead lock prevention and avoidance.

12. Describe memory management process with relevant examples.

13. Write detailed notes on the following :

    (a) File storage

    (b) Directory structure.

14. Write an overview about paging concepts.

15. How to analyze swapping system? Explain.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

RELATIONAL DATABASE MANAGEMENT SYSTEM

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Discuss the features of DBMS.
2. Brief on different User Interfaces.
3. Explain the concepts of Relational Model.
4. Write short notes on Data control facilities of Relational model.
5. Explain the different types of databases.
6. Write short notes on Oracle users.
7. Discuss on Database objects.
8. Explain Database Index in Oracle.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Explain Database Schema and Subschema.
10. Describe the mapping operation in Relational model.

11. Detail on Data Manipulation facilities in Relational model.

12. Describe the Oracle Database Architecture.

13. Explain the following:

    (a) Oracle Queries

    (b) Multipart Queries.

    (c) Joining multiple tables in a Query.

14. Write and explain the commands with their syntax to do the following:

    (a) Modifying table

    (b) Copying a table

    (c) Deleting a table.

15. Explain in detail on Embedded SQL statements.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

COMPUTER NETWORKS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Distinguish between TCP/IP and OSI reference model.

2. Explain the working principle of telephone system with a neat diagram.

3. Write note on ALOHA protocol.

4. Explain carrier sense multiple access protocol.

5. Elaborate on tunnelling.

6. Write short notes congestion control algorithm.

7. Discuss the service primitives of transport layer.

8. Write short notes on DNS.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the various transmission media mention the advantages of each of them.

10. Discuss ISDN and its types.

11. Explain error detection and error correction codes.

12. Write short notes on :

    (a) HDLC.

    (b) SLIP and

    (c) PPP.

13. Discuss various internet control protocols.

14. Explain internet transport protocol.

15. Write about data compression techniques.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

INTERNET AND JAVA PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on :

   (a) Domain Name System

   (b) Search Engines.

2. Give a detailed note on classes and objects.

3. Define array. Explain with an example.

4. Discuss about AWT classes in detail.

5. List out various graphics methods. Explain any three of them.

6. Explain any eight HTML tags with the properties and write web page using that tags.

7. How do you design forms in web application using HTML? Give an example.

8. Write short note on TCP/IP sockets.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) How do you create an E-mail?

   (b) Discuss any two web browsers.

10. Describe the data types available in Java.

11. How exception handling can be performed? Explain with example.

12. Discuss the control structures in Java Script.

13. Write a simple Java Script to develop a web application.

14. Discuss various I/O streams.

15. What are the characteristics of JDBC? Discuss various steps for using JDBC.

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

SOFTWARE ENGINEERING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Brief on prototyping model.
2. Describe the software project planning.
3. Brief on Cocomo model.
4. Describe the characteristics of quality assurance.
5. Brief on requirement analysis.
6. Describe the characteristics of requirement specification.
7. State the guidelines for design the software quality.
8. Discuss about unit, integration and valuation testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Briefly explain the role of management in software development.

10. Explain the resources required for software project planning.

11. Explain the task set for project scheduling.

12. Explain the software configuration management principles.

13. Describe the requirement specification.

14. Explain the modularity in design.

15. Explain the software testing methods.

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

COMPUTER GRAPHICS AND MULTIMEDIA

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain the Bresenham's line drawing algorithm in 2D. Give the pixel positions for the line joining the points (4, 4) and (9, 9).

2. Explain the Cyrus-Beck algorithm for clipping a line with respect to a rectangular window.

3. What are hidden lines and surface? Explain the mechanism of removing hidden surfaces.

4. What do you mean by hidden lines and surfaces? Describe area subdivision method for removing hidden surfaces.

5. Describe rendering pipeline in OpenGL.

6. Define Hypermedia. Links, Anchors and Nodes.

7. Describe the JPEG compression scheme.

8. Explain where and why the magnetic storage is used in multimedia system?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write the midpoint circle drawing algorithm and explain how it works with an example.

10. Discuss in detail about Bezier Curve and surfaces.

11. (a) Describe video compression methods, what are the steps to optimize video files on CD-ROM.

    (b) Describe briefly the differences between MIDI and digital audio

    (c) What is 'Hypertext"? Explain how it helps in presentation and use of information?

12. Describe Phong shading model for rendering polygon surface. How is it better than Gourad shading model? How can you make Phong shading model fast?

13. What do you mean by authoring tool? Describe the grouping of the authoring tools.

14. What are the main design issues for the multimedia authoring system? Briefly explain each one of these issues.

15. Discuss about the application and transport subsystems in multimedia communication systems.

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DISTANCE EDUCATION

M.Sc. (Software Engg.) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

UNIX AND SHELL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain the architecture of UNIX with the help of a neat diagram.

2. Discuss about the following commands :

   (a) Comm

   (b) Cmp

   (c) Diff.

3. Write short notes on pre defined variables.

4. Discuss about pipes.

5. Write about sed with an example.

6. Explain about user-defined functions.

7. Mention the Korn Shell features.

8. Write short notes on system calls.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Explain briefly about the file handling utilities of UNIX. (10)

(b) Discuss about any five networking commands. (5)

10. (a) Write a UNIX program to compare two files. (10)

(b) Discuss about filters. (5)

11. (a) Explain briefly about the various string functions with suitable examples. (12)

(b) Write short notes on awk. (3)

12. Discuss briefly about decision making statements with suitable examples. (15)

13. Write short notes on :

(a) Selection statements. (8)

(b) Repetition statements. (7)

14. Write an essay on file management in UNIX. (15)

15. (a) Discuss about directory API. (8)

(b) Write short notes on special parameters and variables. (7)

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

VISUAL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe the environment of VB.

2. With neat diagram explain about menus.

3. Discuss about VB and the internet.

4. Write a program in VC++ to multiply by matrices overloading operator.

5. Explain MFC file handling in detail.

6. Explain the mouse and keyboard events.

7. Explain the Splitter Windows

8. Develop a database application for maintaining student details

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Compare and contrast between windows programming and traditional programming model.

10. Explain about displaying and printing information in VB.

11. Discuss in detail about data control with an example program.

12. Explain in detail about the document view architecture.

13. Explain about reading and writing documents using MDI applications.

14. What is DLL? Explain its types and advantages in detail. write a program to create a user defined DLL.

15. Explain about the object linking and embedding.

DE–2861

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

SOFTWARE PROJECT MANAGEMENT AND METRICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write note on cost budgeting and cost control.

2. Discuss about the peer inspections.

3. What are the seven workflows in the lifecycle?

4. What levels of activity take place in these workflows during each of the two phases? (Inception and transition).

5. Discuss project interaction management.

6. Write a detailed essay on software configuration management.

7. Why are the metrics divided into management and quality indicators? Name the core metrics under each category.

8. Discuss about the tailoring the process.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the steps to defining a software development process in detail.

10. What are the components of a good cost estimate, in practice?

11. Explain the following :

    (a) Management artifacts

    (b) Programmatic artifacts.

12. How milestones can be identified in project? Discuss briefly.

13. Explain in detail about the cost and schedule estimation.

14. List the seven core metrics, their purpose and perspectives.

15. What were metrics collected in CCPDS-R? What is the purpose of each metric?

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DISTANCE EDUCATION

M.Sc. (5 Years Integrated) (Software Engineering) DEGREE EXAMINATION, DECEMBER 2015.

WEB TECHNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write an HTML script to demonstrate lists and frames.

2. Discuss the merits and demerits of XML. Compare and contrast with HTML.

3. Discuss about BDK.

4. Describe in detail about web server.

5. Write the anatomy of JSP Page.

6. Write short notes on JDBC.

7. How to execute Servlet?

8. What are all the steps involved in JSP-MS access Connectivity?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. How will you display an XML document? Explain.

10. Write shorts on (a) EJB (b) Java beans API.

11. Discuss the applications of Servlets.

12. Discuss in detail about action elements in JSP with an example of display current time and color.

13. Explain how to deploy java beans in JSP page.

14. Explain the following :

    (a) servletContextListener

    (b) servletSessionListener

    (c) SessionAttributeListener

    (d) SessionObjectBindingListener

    (e) RequestAttributeListener

15. Explain different forms of HTML. Illustrate with examples.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

DATA WAREHOUSING AND MINING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Compare data mining and query tool.

2. Discuss the major issues in data mining.

3. Explain how to apply neural networks in discovery of classification rules.

4. Write short notes on Meta data.

5. Explain briefly about association rules.

6. Discuss about K-means algorithms.

7. What is WEB data? Discuss briefly about it.

8. Discuss about Genetic algorithms.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Mention and explain the applications of data mining.

10. Discuss in detail about knowledge discovery process.

11. Briefly explain about the impact of outliers on clustering.

12. Explain the following :

    (a) Categorical Algorithm

    (b) KNN

13. Describe Web content mining in detail.

14. Explain in detail about any two clustering algorithm.

15. What is OLAP? Discuss about the impact of OLAP in data mining.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

DISTRIBUTED COMPUTING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is distributed system? Discuss the advantage of it.

2. Explain how remote invocation is used in distributed system.

3. Write notes on digital signatures.

4. List the difference between event and process.

5. What is concurrency control? Explain how it is helpful in distributed environment.

6. Write notes on RPC.

7. What is distributed deadlock? Discuss it.

8. Write notes on DNS.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the various network technologies.

10. Discuss about the client server communication.

11. Explain in detail about sun network file system.

12. Discuss briefly about the multicast communication related problems.

13. Describe the distributed multimedia systems in detail.

14. Write notes on :

    (a) Processes and threads

    (b) Global name service.

15. Explain the cryptographic algorithms in detail.

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DISTANCE EDUCATION

M.Sc. (Five Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

Software Engineering

SOFTWARE TESTING AND REUSE

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is software testing? Explain the purpose of testing in detail.

2. Write short notes on data flow testing.

3. List the difference between nice and ugly domains.

4. Write short notes on reduction procedure.

5. What is KV chart? Discuss about it.

6. What is state graph? Why it is needed?

7. Briefly explain the kinds of testing on a typical software system.

8. What is rapid testing? List the advantage of it.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss in detail about the model for testing.

10. Explain the transaction flow testing techniques.

11. Briefly discuss about path products and path expressions.

12. Describe about good state graph and bad state graph.

13. Explain the usage of win-runner tools in testing.

14. Write notes on :

    (a) Decision tables

    (b) State testing

15. Discuss about the powers of a matrix in graph matrices.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

CRYPTOGRAPHY AND NETWORK SECURITY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain various security attacks.

2. Explain briefly about key distribution.

3. Explain secure hash function.

4. Explain directory authentication service.

5. Write short notes on digital certificates.

6. Explain IP security architecture.

7. What is PGP? Explain briefly.

8. Write short notes on 'Firewall'.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain any two security services.

10. Describe the DES algorithm.

11. Explain the RSA algorithm.

12. Explain TCP session hijacking and UDP hijacking.

13. Explain encapsulating security payload.

14. Briefly explain about secure socket layer (SSL) and Transport Layer Security (TLS).

15. Explain the trusted systems.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

SOFTWARE QUALITY ASSURANCE AND STANDARDS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is Software Quality? Explain the traditional waterfall lifecycle model with neat sketch.

2. Explain the four quality attributes of Gilb's approach.

3. Discuss about software quality assurance plan.

4. What are the advantages of CASE tools? Explain the types of CASE tools.

5. Explain the Rayleigh Model.

6. What are the main elements of Quality Management System? Write notes on each of the elements.

7. Explain about customer satisfaction analysis.

8. Write down the purpose of standards. List out the ISO 9000 series of quality management standards.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the hierarchical models of Boehm and Mccall with neat sketch.

10. Discuss in detail about software metrics and explain the relationship between criteria and measurable properties.

11. (a) Write notes on characteristics of software quality assurance.

    (b) Write about SQA reviews and audits.

12. Explain about perfect prevention and removal.

13. Discuss about reliability growth models.

14. Explain the complexity metrics and quality management models.

15. Discuss in detail about capability maturity model.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

MOBILE COMMUNICATIONS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Briefly discuss the GSM technology.

2. Explain about the cyclic repetition of data.

3. Describe about IEEE 802.11.

4. Write short note on Bluetooth technology.

5. Explain the method of IP packet delivery.

6. Discuss in detail about IPV6.

7. Explain about snooping TCP.

8. Write short note on session protocol.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write an overview of broadcast system.

10. Explain in detail about the access point control protocol.

11. Discuss about tunneling and encapsulation.

12. Explain in detail about Adhoc networks.

13. Explain about transaction oriented TCP.

14. Describe the architecture of the transaction protocol.

15. Explain briefly about wireless telephony application.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

OPEN SOURCE ARCHITECTURE

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is PAM authentication? Discuss.

2. Briefly discuss on the basics of TCP/IP networking and routing.

3. Write a short note on OS boot up process.

4. How do you setup printer services? Discuss.

5. How do you use CVS to manage source code revisions? Discuss.

6. Discuss the construction of make files and the usage of make files.

7. Write a short note on the C preprocessor.

8. Discuss the basics of the X windows server architecture.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. How do you connect to the Internet through dialup, DSL, Ethernet, leased line? Discuss.

10. Give an overview of free/open source software.

11. How do you perform every day tasks using gnu/Linux?

12. Discuss briefly on setting up a firewall.

13. Write a short note on the usage of GNU compiler collection.

14. Give a brief note on bash, sed and awk scripting.

15. Write a brief note on Gtk+ programming.

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DE–9073

Sub. Code
16

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

C PROGRAMMING LAB

Time : Three hours Maximum : 100 marks

ONE question to be selected by Lot system.
Each sub-division carries 50 marks.

1. (a) Write a C program to find the roots of a quadratic equation using function definition.

   (b) A file VOCABULARY contains 10 words. Write a program to write in another file WORDS, the words beginning with the character S or s from the 10 words.

---------- Cut here ----------

2. (a) Write a C program using functions to replace a substring in a string and to find the length of a string.

   (b) Write a C program to sort the given numbers in ascending order using pointer.

---------- Cut here ----------

3. (a) Write a C program to arrange the names in alphabetical order and assign roll number to each student.

   (b) Write a C program to check whether the given word is palindrome or not.

4. (a) Write a C program for matrix manipulation (addition, subtraction, multiplication) using menu driven.

(b) Write a C program to find the largest and next to largest element in an array.

---------- Cut here ----------

5. (a) Write a program to generate Armstrong numbers from 100 to 999.

(Example: $153 = 1^3 + 5^3 + 3^3$)

(b) Write a program to merge two arrays.

---------- Cut here ----------

6. (a) Write a program to copy a structure of your choice to another structure.

(b) Write a program to count the number of words in a given sentence.

---------- Cut here ----------

7. (a) Write a recursive function to convert a decimal number to binary.

(b) Write a C program to process student records by using structures.

---------- Cut here ----------

8. (a) Write a C program to convert upper case letter to lower case and lower case letters into upper case in a text file.

(b) Write a C program to read 10 values to an array variable. Use pointers to locate and display each value.

DE–9074

Sub. Code
17

DISTANCE EDUCATION

M.Sc. (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

Software Engineering

DATA STRUCTURES LAB

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by LOT system.
Each sub-division carries 50 marks.

1. (a) Write a "C" program to copy contents of one file to another file.

   (b) Write a "C" program to implement insert and delete operations on linked list structure.

---------- Cut here ----------

2. (a) Write a "C" program for linked list implementation of Queue operations.

   (b) Write a menu driven program in "C" to find an element using linear and binary search methods.

---------- Cut here ----------

3. (a) Write a "C" program to implement insert and delete operations on linked list structure.

   (b) Write a "C" program for linked list implementation of Queue operations.

4. (a) Write a "C" program to evaluate the given mathematical expression using stack.

   (b) Write a "C" program to sort numbers in ascending order with naming of and display the values before and after sorting.

---------- Cut here ----------

5. (a) Write a "C" program to implement insert and delete operations on Queue using array concept.

   (b) Write a "C" program to sort 15 nos in descending order and display the values before and after sorting.

---------- Cut here ----------

6. (a) Write a "C" program to implement push and pop operations on stack.

   (b) Write a "C" program to sort a set of elements using selection sort.

---------- Cut here ----------

7. (a) Write a "C" program for linked list implementation of Queue operations.

   (b) Write a "C" program to sort a set of elements using selection sort.

---------- Cut here ----------

8. (a) Write a "C" program to copy contents of one file into another file.

   (b) Write a "C" program to sort a set of elements using insertion sort.

DE–9075

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, DECEMBER 2015.

C++ – LAB

Time : Three hours Maximum : 100 marks

ONE question to be selected by lot system.

Answer both (a) and (b)

1. (a) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

   (b) Using Pointers. Create a class and write a program to get names and display them in sorted order.

---------- Cut here ----------

2. (a) Using operator overloading, write a C++ program to find the difference and total length of given two various tubes specifies in meters and centimeters.

   (b) Write a menu driven C++ program to add and substract given two matrices of order m×n defined in class, using operator overloading.

---------- Cut here ----------

3. (a) Write a program to create a class for representing a "Course", with suitable properties and methods. Also write the main function to create an object to represent the details of your course.

   (b) Write a C++ Program to read an array of numbers, to count the number of elements which are greater than 100 and to display the result by using pointer.

4. (a) Write a C++ program to process the student information through files (with the options for insert, delete, update, display)

(b) Write an interactive C++ program to process complex numbers. It has to perform addition, subtraction, multiplication and division of complex numbers. Print results in $x + iy$ form.

---------- Cut here ----------

5. (a) Using a dynamic constructors, write a C++ program to concatenate two given strings.

(b) Using pointers, create a class and write a program to get n names and display them in sorted under.

---------- Cut here ----------

6. (a) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

(b) Assume you want to generate a table of multiples of any number. Write a program that allows the user to enter the number and then generates the table, formatting it into ten columns and 20 lines.

---------- Cut here ----------

7. (a) Write a program to process marks with the help of classes. All subjects mark must be ≥ 50 for pass otherwise fail.

(b) Write a program using pointers to get a given number of names and display them in ascending order.

8. (a) Write a C++ program to read an array and to find the number of occurrences of a given number.

   (b) Write a program in C++ to display the elder among three persons using 'this' pointer.

DE–9076

Sub. Code
27

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, DECEMBER 2015.

RDBMS – LAB

Time : Three hours Maximum : 100 marks

ONE question to be selected by lot system.

Answer both (a) and (b)

1. (a) Create a table called 'Product' with the following details:

Fieldname	Field type	Field size
Product	Character	15
Product Name	Number	25
Qty on hand	Number	20
Cost price (Rs.)	Number	6with2decimalplaces

Queries:

(i) Enter the at least 15 records for the table

(ii) List of all products having 'a' as the second letter in their names.

(iii) List the Product codes and names where the Cost price is greater than Rs.100/-

(iv) List the Product codes where the quantity on hand is more than 50.

(v) List the Product Name, Qty on hand, Cost price and the Total cost price of each product.

(b) Write a PL/SQL program to split the SSLC Marksheet table into two tables with one containing the names of passed students and the other containing the names of failed students.

2. (a) Create a table Employee with the fields like empname, salary, dept, and designation.
   (i) List the name of the Manager
   (ii) List the name of the entire employee other than Manager.
   (iii) Find the sum of the salary of all employees.
   (iv) Find Max and Minimum salary of all employees.

   (b) Write a PL/SQL program to display names of the employees in the Employee table whose names are palindrome.

---------- Cut here ----------

3. (a) Write a SQL program Electricity billing system with the following fields meterno, cusname, LMR (Last month Reading), CMR (Last month Reading), consumed.
   (i) List the customer name started with "C".
   (ii) List the meter numbers between 100 to 200.
   (iii) If consumed units is less then 200 amount is Rs.150/-.
   (iv) If consumed units between 201 and 300 cost per unit is Rs.3/-
   (v) If consumed units between 301 and 400 cost per unit is Rs.5/-
   (vi) Else Rs.10/-.

   (b) Create a oracle table with the table name as EMP with following :

   fields:(s.no, empID, Name, Desgination, DeptID, DeptNAme, Basicsal, DA, HRA, IC, NETSAL, GROSSSAL)
   (i) Find NETSAL and GROSSSAL
   (ii) Find out the details of employees whose names begin with 'S'.
   (iii) Compute yearly salary of anyone employee
   (iv) Calculate the total monthly salary of all employees
   (v) Print the average annual salary.

4. (a) Create a table Employee with the fields like empname, salary, dept, salary, designation.
   - (i) List the name of the Manager
   - (ii) List the name of the entire employee other than Manager.
   - (iii) Find the sum of the salary of all employee
   - (iv) Find Max and Minimum salary of all employee.

   (b) Create a table sales_order_details with the following fields s_order_no product_no, description, qty_ordered, qty_disp,product_rate, profit_percent, sell_price, supplier_name.
   - (i) Select each row and compute sell_price*. 50 and sell_price*1.50 for each row selected.
   - (ii) Select product_no, profit_percent, Sell_price where profit per is not between 10 and 20 both inclusive.
   - (iii) Select product_no, description, profit_percent, sell_price where profit_percent is not between 20 and 30.
   - (iv) Select the suppliername and product_no where suppliername has 'r' or 'h' as second character.

---------- Cut here ----------

5. (a) Create a oracle table with the table name as STUDENT
   - (i) INSERT the values in the table
   - (ii) Display the students name who's Rollno is 5
   - (iii) Display the students percentage between 70 And 85
   - (iv) UPDATE the PERCENTAGE=90 WHERE Rollno=3;
   - (v) Delete the table

(b) Write a SQL program Electricity billing system with the following fields meterno, cusname, LMR (Last month Reading), CMR (Last month Reading), consumed.

(i) If consumed units is less then 200 amount is Rs.150/-

(ii) If consumed units between 201 and 300 cost per unit is Rs.3/-

(iii) If consumed units between 301 and 400 cost per unit is Rs.5/-

(iv) Else Rs.10/-

DE–9077

Sub. Code
36

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

INTERNET AND JAVA – LAB

Time : Three hours Maximum : 100 marks

One questions to be selected by lot system Answer both (a) and (b).

1. (a) Write a Java script to create Order Form to select the house articles.

   (b) Write a Swing program to create the Tabbed panels.

---------- Cut here ----------

2. (a) Write a Java program using Applet to create the frames and its control.

   (b) Write a Java program using applet to display any 3 images when 3 buttons in the border layout are clicked. The image should be displayed at the center.

---------- Cut here ----------

3. (a) Write a Menu-Drive java program to do database functions using database connectivity facility. The functions should display, edit, insert and delete a record from a table.

   (b) Develop a picture gallery having at least 5 pages each of them is having several pictures with suitable information.

4. (a) Write a Java script to create a Color palette and display the text in the color chosen from the palette with proper background color.

(b) Write a Java program to create a color background.

---------- Cut here ----------

5. (a) Write a Java program using Applet to display the different colors and fonts.

(b) Write a Swing program to create Buttons with
(i) Tool tip text

(ii) Image

(iii) Border

(iv) Short cut key.

---------- Cut here ----------

6. (a) Write a VBScript to create a calendar for given month and year.

(b) Using atleast 20 HTML tags to create a screen with a string "WEB Design".

---------- Cut here ----------

7. (a) Write a Java program using applet to display the dialogue menu.

(b) Design and publish a web page for a college.

8. (a) Write a VBScript to do the following:

   (i) Check 'the given password

   (ii) Change the existing password

   (b) Write a Java program using Applet to create the frames and its control.

DE–9078

Sub. Code
37

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

UNIX & SHELL PROGRAMMING – LAB

Time : Three hours Maximum : 100 marks

(i) One question should be given to each candidate by lot.

(ii) Each sub division carries 50 marks.

1. (a) Write a menu driven shell program for the following:

   (i) List of files

   (ii) Processes of users

   (iii) Today's date

   (iv) Users of system

   (v) Quit to UNIX

   (b) Write a shell program to prepare electric bill for domestic consumers

   For first 100 units - Rs. 0.75/unit

   For next 100 units - Rs. 1.50/unit

   Above 200 units - Rs. 3.00/unit

   Prepare the Electricity bill in the following format

   Customer No : _____

   Customer name : _____

   Previous reading : _____

   Current reading : _____

   Units consumed : _____

   Charge : _____

2. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

   (i) File existence

   (ii) File readable

   (iii) File writable

   (iv) Both readable and writable.

   (b) A hospital maintains patient details. Write a shell program to list the following

   (i) Patients in a blood groupwise.

   (ii) Patients in age between 20 to 30.

   (iii) Exit.

---------- Cut here ----------

3. (a) Write a shell program using 3 arguments to take the pattern as well as input and output file names if the pattern is found, display "pattern found" else display "Error message". Also check if right no of arguments are entered.

   (b) Write a program for four – function calculator by using yacc command.

---------- Cut here ----------

4. (a) Write a shell program which periodically monitors the disk for the existence of a file and then Executes the program once the file has been located (using UNTIL statement)

   (b) Write a shell program to sum up the series $\frac{1}{1!}-\frac{1}{2!}+\frac{1}{3!}+.....\frac{1}{10!}$.

5. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it:

   (i) Enter the five names in a file

   (ii) Sort the names in existing file

   (iii) List unsorted and sorted file

   (iv) Quit

   (b) Write and execute a shell program to print monthly pay slips for employees in an organization. Assume your own data and generate accordingly.

---------- Cut here ----------

6. (a) Write a menu driven shell program to copy, edit, rename, delete a file.

   (b) Write and execute a shell program to do the following

   (i) Convert given string in center of given row in BOLD, normal or reverse video.

   (ii) Write given string at given row, column in BOLD, normal and reverse video.

   (iii) Search a given string using awk with various options.

---------- Cut here ----------

7. (a) Write a menu driven shell program to perform the following task.

   (i) Enter the sentence in a file

   (ii) Search a given whole word in an existing file

   (iii) Quit.

   (b) Write and execute a shell program to send mail to groups of users by extracting their name from the group file.

8. (a) A hospital maintains patient details. Write a shell program to list the following

   (i) Patients in a blood groupwise

   (ii) Patients in age between 20 to 30

   (iii) Exit

   (b) Write and execute a shell program to generate leave status report of an organization.

DE–9079

Sub. Code
46

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 years integrated) DEGREE EXAMINATION, DECEMBER 2015.

VISUAL BASIC AND VC++ — LAB

Time : Three hours Maximum : 100 marks

ONE question to be selected by lot system.

Each subdivision carries 50 marks.

1. (a) Write an event procedure for the following :
   (i) Reverse a string.
   (ii) Determine whether the given string is Palindrome or not.
   (iii) Change the case of a string to upper of lower.

   (b) Using MSFlex grid control display the multiplication and addition table of 20 rows and columns (Use MSFlex, Grid, Button).

---------- Cut here ----------

2. (a) Write event procedure for the following :
   (i) Display data and time in the label box at run time.
   (ii) Find the power of a number (Accept two inputs as number and power).
   (iii) Temperature conversion (From Fahrenheit to centigrade) (Use Label, Text box, Command Button).

   (b) Write a visual C++ program to create a list box in window.

3. (a) Write a Visual C++ program to create a window of desired size using MFC.

(b) Using built in AX control, develop the windows NOTE PAD with file, edit menu operation and also display the floating menu whenever necessary. (Use RichTextbox, Menu editor).

---------- Cut here ----------

4. (a) Using a control array, create a simple calculator which will do the following operation

(i)	Addition	(ii)	Subtraction
(iii)	Multiplication	(iv)	Division
(v)	Square	(vi)	Square root
(vii)	Modulus	(viii)	Power
(ix)	Percentage.		

(Use Textbox, Comment Button, Frames)

(b) Write a VC++ program to find out whether a mouse is attached or not ; and if attached how may buttons are present or not.

---------- Cut here ----------

5. (a) Write a VC+++ program to handle windows messages in MFC program.

(b) Create a worksheet with the following fields Student Name, Internal Mark. External Mark and Result.

Using data access object, create a student application to do the following operations.

(i) Insertion (ii) Deletion (iii) Updation.

6. (a) Write a VC++ program to get the status of the shift and toggle keys using MFC.

   (b) Write a VB program to perform the following operation in a record of random Access file

   (i) Insert (ii) Delete (iii) Search.

---------- Cut here ----------

7. (a) Create a VB application with a Drive List Box, Dir List box

   (i) Select an exe file and execute it.

   (ii) Select a picture and load it to the form.

   (iii) Fitter the file in the file list box according to the extension chosen in the Combo box.

   (b) Write a VC++ program to fill background of the client area with a bitmap.

---------- Cut here ----------

8. (a) Write a VC++ program to generate a status bar and show the status of Caps Lock, Nom Lock and Scroll Lock in it.

   (b) Create a form with Text box, Combo box and Command Button to do the following operation

   (i) Add the University Name in the Combo box at the run time.

   (ii) Search and delete the particular University.

   (iii) Display the message box with available number of University.

DE–9080

Sub. Code
47

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

WEB TECHNOLOGY — LAB

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Create a HTML table with rows and columns and split them using row span and col span.

   (b) Write an ASP program to prepare student performance evaluation document using Java Script.

---------- Cut here ----------

2. (a) Create a web page in the format of front page of a newspaper using text links. Align the text with colors.

   (b) Develop an application for JSP-servlet communication.

---------- Cut here ----------

3. (a) Create an image animation using Java Script.

   (b) Write an XML document to display your bio-data. Write an XSL style sheet and attach that to the XML document.

4. (a) Write a servlet program to display "Hello world".

   (b) Write a JSP program to display student mark list using JDBC.

---------- Cut here ----------

5. (a) Write an ASP program to prepare student performance evaluation document using Java Script.

   (b) Develop an application for JSP servlet communication.

---------- Cut here ----------

6. (a) Create a web page in the format of front page of a newspaper using text links. Align the text with colors.

   (b) Write servlet program to send E-mail message.

---------- Cut here ----------

7. (a) Write an ASP program to prepare employee pay bill using Java Script.

   (b) Create a HTML table with rows and columns and split them using row span and col span.

---------- Cut here ----------

8. (a) Write a servlet program to send E-mail message.

   (b) Write an XML document to display your bio-data. Write an XSL style sheet and attach that to the XML document.

DE–9081

Sub. Code
56

DISTANCE EDUCATION

M.Sc. (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

Software Engineering

OPEN SOURCE LAB

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

Each sub division carries 50 marks.

1. (a) Write a server side PHP program that displays marks, total, grade of a student in tabular format by accepting user inputs for name , number and marks from a HTML form.

   (b) Write a PHP program that adds products that are selected from a web page to a shopping cart.

---------- Cut here ----------

2. (a) Write a PHP program interface to create a database and to insert a table into it.

   (b) Write a PHP program to access the data stored in a my sql table.

3. (a) Write a PHP program using classes to create a table.

(b) Write a PHP program to upload a file to the server.

---------- Cut here ----------

4. (a) Write a PHP program to create a directory and to read contents from the directory.

(b) Write a shell program to find the details of an user session.

---------- Cut here ----------

5. (a) Write a shell program to change the extension of a given file.

(b) Create a my sql table and execute queries to read, add, remove and modify a record from that table.

---------- Cut here ----------

6. (a) Write a PHP program to access the data stored in a my sql table.

(b) Write a PHP program to upload a file to the server.

---------- Cut here ----------

7. (a) Write a shell program to find the details of an user session.

(b) Write a PHP program using classes to create a table.

---------- Cut here ----------

8. (a) Write a shell program to change the extension of a given file.

(b) Write a PHP program to access the data stored in a my sql table.

DE–9082

Sub. Code
57

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2015.

Software Engineering

CASE TOOLS LAB

Time : Three hours Maximum : 100 marks

One question to be selected by lot system.

Each subdivision carries 50 marks.

1. (a) Perform program analysis and project planning to illustrate project scope identification, objectives and infrastructure.

   (b) Develop a student mark analysing system and perform validation testing.

---------- Cut here ----------

2. (a) Develop a software that illustrate various modules of a project and identifies deliverables.

   (b) Develop a Quiz system and perform converage analysis and memory leaks.

---------- Cut here ----------

3. (a) Develop a software that illustrates the way of building and testing class diagram.

   (b) Implement an online ticket reservation system and perform validation testing.

4. (a) Develop an application software that illustrates the way of debugging.

   (b) Implement a Payroll system and develop test case hierarchy site check and site monitor.

---------- Cut here ----------

5. (a) Develop an expert system and perform coverage analysis.

   (b) Develop a Real-time scheduler and develop test case hierarchy site check and site monitor.

---------- Cut here ----------

6. (a) Perform program analysis and project planning to illustrate project scope identification, objectives and infrastructure.

   (b) Develop an ATM system and perform validation testing.

---------- Cut here ----------

7. (a) Perform software requirement analysis to describe the modules of a project and to identify deliverables.

   (b) Develop course registration system and perform coverage analysis.

---------- Cut here ----------

8. (a) Develop a software to illustrate the method of debugging.

   (b) Develop a software for stock maintenance and perform memory leaks analysis.